Position Vectors

Exam code: 9MA0

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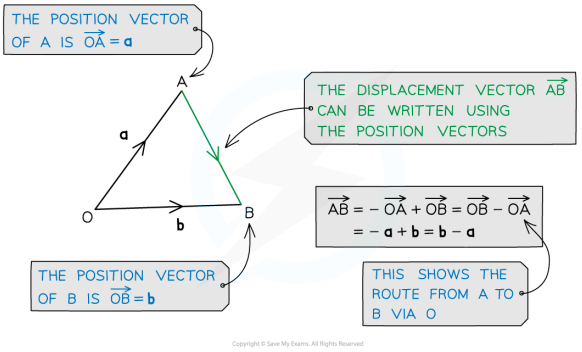
Help me understand

Position vectors

What is a position vector?

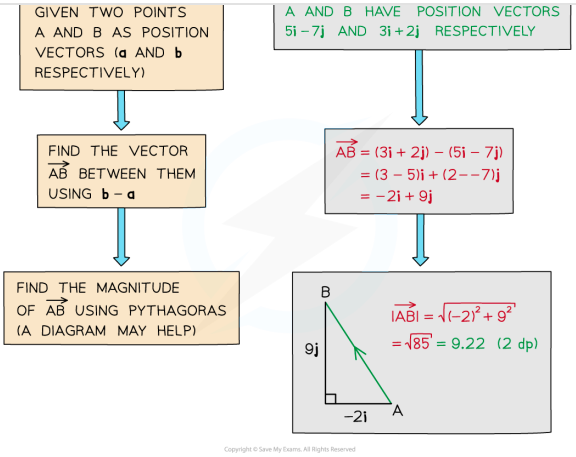
- Position vectors describe the **position of a point** in relation to the origin
- They are different to **displacement vectors** which describe the **direction and distance** between any two points



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Distance between two points

- The distance between two points is the **magnitude** of the vector between them (see Magnitude Direction)



How do I find the magnitude of a displacement vector?

- You can use **coordinate geometry** to find **magnitudes** of **displacement vectors** from A to B
  - From the **position vectors** of A and B you know their coordinates
    - If $\mathbf{a} = \overrightarrow{OA} = x_1\mathbf{i} + y_1\mathbf{j} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$, then point A has coordinates (x_1, y_1)
    - If $\mathbf{b} = \overrightarrow{OB} = x_2\mathbf{i} + y_2\mathbf{j} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$, then point B has coordinates (x_2, y_2)
  - The **distance** between two points is given by $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
    - So $|\overrightarrow{AB}| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
  - For example, if points A and B have position vectors $5\mathbf{i} + 3\mathbf{j}$ and $3\mathbf{i} - 6\mathbf{j}$ respectively
    - then $|\overrightarrow{AB}| = \sqrt{(5-3)^2 + (3-(-6))^2} = \sqrt{85} = 9.22 \text{ (3 s.f.)}$
- Alternatively, you could find $|\overrightarrow{AB}|$ by
  - first using $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB}$ to find $\overrightarrow{AB}$ in **vector form**
    - and then calculating its **magnitude** directly
  - See the Worked Example below

Examiner Tips and Tricks

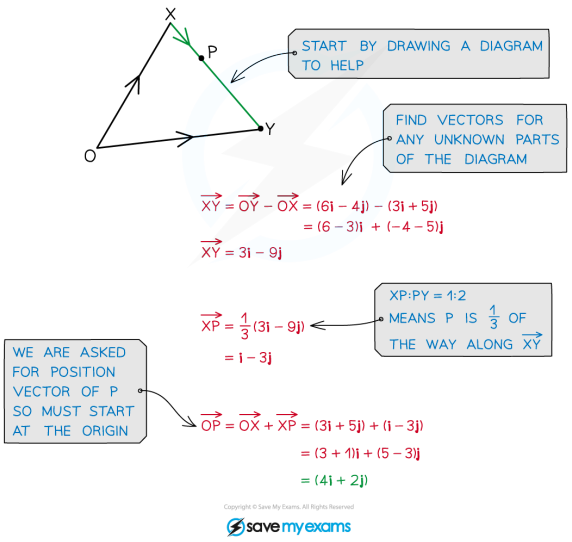
- Remember if asked for a position vector, you must find the vector all the way from the origin.
- Diagrams can help, if there isn't one, draw one.

Worked Example

The points X and Y have position vectors $3\mathbf{i} + 5\mathbf{j}$ and $6\mathbf{i} - 4\mathbf{j}$ respectively.

Point P is on the line XY such that XP : PY = 1 : 2.

Determine the position vector of P.



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